Course: CSE 3523 Microprocessor, Microcontroller and Embedded Systems

Semester: 5th (Autumn_25)

Guidance for Final Exam Preparation

Segment 4: Interrupt System

Interrupt definition, Interrupt types, Interrupt handling in 8086, Interrupt vector table, IVT format and example, Interrupt types (0–255), Type 0 (divide-by-zero), Type 1 (single-step), Type 2 (NMI), NMI examples, Type 3 (breakpoint), Type 4 (overflow), , Type 5 (BOUND check),Compariaion Table of Type 0-5, IRET instruction, Interrupt sources, NMI power failure circuit, INTR–INTA– Data bus, Fixed interrupt vector gen, DIP switch with 74ALS244, INTR expansion (74ALS244+ NAND), Daisy chain with 82C55

Segment 5:

Multitasking concept, Multi-user systems, Coprocessor, Dual-core, Cache memory, Cache types (L1–L3), Instruction Execution Types ,non pipelining,pipelining:key points,pipelining stages,arithmatic pipeline,Instruction pipeline,pipeline Hazards,Advantages and disadvantages,Specialized processors: Embedded type, Bit-slice type, Arithmetic type

Segment 6: I/O Operation

IN/OUT instructions, Fixed vs variable I/O addressing, Isolated vs memory-mapped I/O, Basic input using 74ALS244, Basic output using 74ALS374, Minimum mode memory interface, 8086 memory org, Address decoding, NAND gate decoder, 3-to-8 decoder (74LS138), Memory expansion (32K×16, 64K×16), 82C55 PPI ports,diagram, 82C55 operation mode 0, Mode 1,Mode 2, Command byte structure, Handshaking concept, Mode 1 input, Mode 1 output.

Segment 7(part-1): 8051 Microcontroller.

Microcontroller, microcontroller vs General purpose microprocessor, Application of 8051,8051 features, Architectural block diagram, ALU, Accumulator, B register, PC, DPTR, I/O ports, timers, interrupts, oscillator, serial communication, pin diagram and functions (Port 0–3, EA, ALE, PSEN), RAM structure (banks, bit-addressable, stack), ROM organization, EA pin role, PSW and flag bits, SFR overview, interrupt sources and handling.Assembly basics, addressing modes,

arithmetic (ADD, ADDC, MUL, DIV), branching (SJMP, JZ, DJNZ), bit-level control (JB, JNB, JBC), assembler directives (ORG, EQU, DB), sample programs (Program 1, Program 2, Program 3, Program 4, Program 5) 7-segment interfacing , Interfacing DC motor with 8051,

Segment 7(Part-2): Embedded Systems & FSM Design

Embedded system definition, types (small, medium, complex), core elements (HW, SW, RTOS), design steps (Req,spec, arch, component, integration,), Application of embadded system,traditional vs HW/SW co-design, Steps of Co Design-Coputational Model, DFG,CDFG,FSM modeling examples (seatbelt, vending matchine, elevator),

Segment 8: IoT and its Technology

Introduction to IoT: Core Concepts and Technology Fundamentals(page-1) IoT Architecture: Core Components & Technology Fundamentals(page-2), Sensors & Actuators: Fundamental IoT Technology(page-3), IoT Infrastructure, Network Architecture & Communication Protocols(page-4), IoT Data Management Systems: Processing for Value Creation(page-5), User Interfaces for Diverse IoT Applications(page-6**),** From M2M to IoT: Evolution and Value Chains(page-7) , Value Creation from Big Data & Serialization (page-8),IoT Applications Across Domains(page-9) Security & Privacy: The Paramount Challenge in IoT(Page-10), Building Trust and Aggregating Data in
Connected Ecosystems(page-11)